Kinetic surrogates for techno-economic screening of ethane cracking

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Abstract

Ethane-cracker operating conditions determine both ethylene yield and the gas flow processed per tonne of product. We quantify how heating history changes process loads, costs, and carbon emissions by integrating detailed kinetics with an open-source process model. At 850 °C outlet temperature and 0.35 s residence time, delaying heating over the tested profile range increases calculated fired-case cost from 456 to 806 USD t^{-1} under fixed process and price assumptions. In the temperature–residence-time scan, longer residence beyond the maximum-yield condition lowers both ethylene yield and total production cost: lower heating cost and higher coproduct credits outweigh additional feed and processing costs. Direct Cantera calculations confirm this ordering, and comparisons with three published process cases quantify the downstream consequences of composition differences. With the same reactor output and equipment costs, electric heating becomes cheaper than fired heating below 2.28 cents/kWh, assuming natural gas at 4 USD/GJ and heating efficiencies of 95% and 60%, respectively.

1 Introduction

Ethane cracking couples high-temperature reaction chemistry to energy-intensive product recovery. Changes in conversion and selectivity alter the amounts of unreacted ethane and coproducts that must be compressed, chilled, and separated per tonne of ethylene. Chen et al. illustrate the importance of this downstream burden: their oxidative-dehydrogenation process has a lower minimum selling price than steam cracking but higher life-cycle emissions, with refrigeration contributing 1.30 kg CO_2e per kilogram of ethylene[1]. Shin et al. show that the economics and emissions of cracker electrification depend on electricity supply and the treatment of hydrogen-rich coproducts[2]. Shortcut column models and separation-network synthesis provide methods for evaluating composition-dependent recovery loads during process design[4,5].

Internal electric heating provides additional control over heat delivery. Balakotaiah and Ratnakar analyze resistance-heated reactor configurations in terms of electrical supply, transport, and reaction time scales[7]. Implementations include catalyst-coated SiSiC foams for methane reforming, FeCrAl coils for catalytic ethane dehydrogenation, and induction-heated ceramic foams for reverse water-gas shift[8–10]. These configurations place heat generation within the reactor, with different solid-to-gas heat-transfer and loss mechanisms. Carbon-fiber-paper reactor calculations resolve localized heating and buoyancy-driven flow[11], while material screening quantifies how electrical and thermal properties constrain heating rates and temperature uniformity[12]. Mittal et al. combined kinetic calculations, transport modeling, and carbon-fiber-paper experiments to examine short-contact-time ethane cracking. Their process analysis linked the predicted product distribution to

energy demand and ethylene cost, with the economic outcome depending on electricity supply[13].

Reactor-to-process integration also underlies the comparison by Cattry et al., who combine internally resistance-heated and fired reactor models with process simulation, economics, and life-cycle assessment[3]. Their reactor design with the highest ethylene yield differs from the minimum-cost design because added heating intensity increases electricity demand. A recent steam-cracker retrofit preprint similarly combines kinetics, process simulation, and optimization to evaluate olefin yield and coproduct value[6]. These studies establish the need to assess reactor improvements against the cost of the full process. Here, we examine how the time spent at high temperature changes fresh-feed demand, ethane recycle, separation loads, and production cost. The comparisons retain the same downstream correlations and price assumptions so that these consequences can be traced to the reactor outputs.

Surrogate models make repeated evaluation of detailed kinetics practical. Bradley et al. describe ways to combine first-principles and data-driven submodels[14], and Döppel and Votsmeier demonstrate surrogate acceleration of surface kinetics[15]. Gaussian-process (GP) models also support computational searches, as implemented in NEXTorch[16]. We use a GP representation of AramcoMech 3.0 calculations in Cantera to examine outlet temperature, residence time, steam dilution, pressure, and heating history[17,18]. Changing the temperature history tests its effects on chemistry and downstream loads, while changing the heat source at the same reactor condition isolates energy substitution. The calculations show how delayed heating increases the flow processed per tonne of ethylene and how heating cost and coproduct credits shift the minimum-cost condition away from maximum yield. Literature-based parameter calibration quantifies how assumed utility discrepancies affect the heating-cost crossover. The released code, training data, and process equations allow users to inspect assumptions and replace individual submodels, including through AI-assisted code review and modification (**Figure 1**).

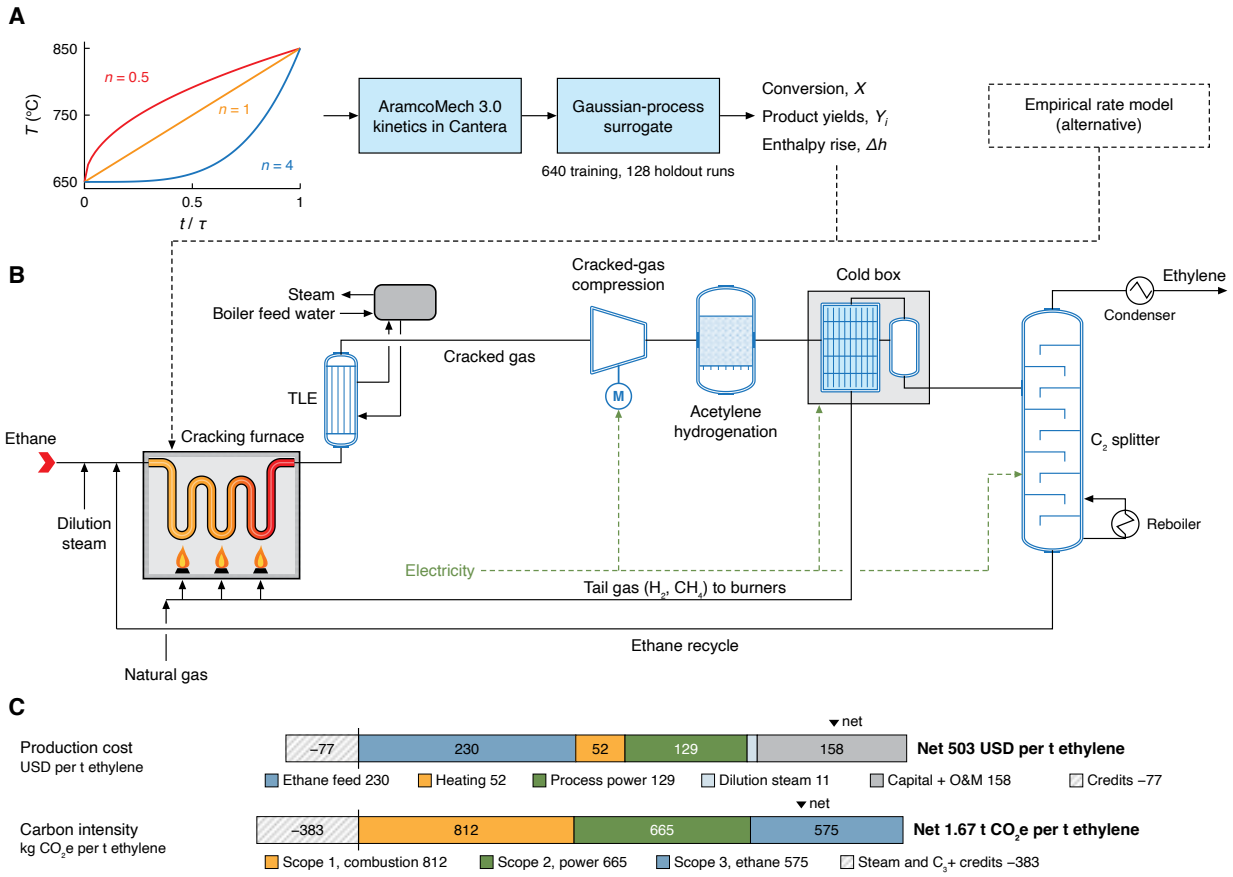


Figure 1. Reaction-to-process screening workflow. **(A)** Prescribed gas-temperature histories at 850 °C outlet temperature, with heating exponents of 0.5, 1, and 4. AramcoMech–Cantera calculations supply conversion, product yields, and enthalpy rise through a GP surrogate trained on 640 calculations and tested on 128 holdout calculations. An empirical rate model supplies the same process interface. **(B)** Process schematic of the fired case showing cracking, transfer-line-exchanger (TLE) heat recovery to steam, cracked-gas compression, acetylene hydrogenation, cold separation, C_2 splitting, and ethane recycle. Tail gas from the cold box fires the burners before purchased natural gas. Electricity drives compression, cold-box refrigeration, and the C_2 splitter. Solid arrows denote material or energy flows, black dashed arrows denote calculation inputs, and green dashed lines denote electricity supply. **(C)** Production cost and carbon intensity by component, per tonne of ethylene. Hatched segments are steam and coproduct credits, and markers show the net values. Values are calculated for the reference fired case at 850 °C, 0.35 s, 1.5 bar, steam/ethane mass ratio 0.35, and heating exponent 1.

2 Methods

2.1 Comparison basis and calculation sequence

The model screens operating conditions and heating options at a common nominal ethylene capacity. Material flows, utility demands, and costs are reported per unit of final ethylene after acetylene conversion. Each case specifies outlet temperature, residence time, steam/ethane ratio, reactor pressure, and gas-temperature history. The kinetic calculation supplies conversion, product yields, and enthalpy rise. The process model then calculates fresh feed, recycle, utility loads, and section costs. Prices, equipment correlations, and product specifications remain fixed unless explicitly varied. Table 1 lists the reference inputs. Run length, the operating time between shutdowns, determines annual production and the fixed charges per tonne.

2.2 Reactor calculations and operating domain

Gas-temperature history is specified as an input so that early and delayed heating can be compared at the same outlet temperature and residence time. The reactor calculation advances a constant-pressure fluid element through this history:

$$\begin{aligned} T(f) &= T_{\text{in}} + (T_{\text{out}} - T_{\text{in}}) f^n \\ f &= t/\tau \end{aligned} \tag{1}$$

Here t is elapsed time and τ is residence time. The inlet temperature T_{in} is 650 °C, and T_{out} is the outlet temperature. The heating exponent n sets the profile shape: $n = 1$ gives a linear temperature rise, $n < 1$ gives earlier heating and a longer period near the outlet temperature, and $n > 1$ delays heating. Product composition is calculated along this path, and the enthalpy change is evaluated from the inlet and outlet compositions and temperatures. Prescribed-temperature studies of periodically operated reaction networks show how exposure time at high temperature controls desired and secondary chemistry[19]. Here, $T(t)$ follows one fluid element during its passage through the reactor.

The training domain covers outlet temperatures of 750–1000 °C, residence times of 0.02–1.5 s, steam-to-ethane mass ratios of 0–0.70, pressures of 1–5 bar, and exponents of 0.45–4. Residence time is sampled logarithmically. Each trajectory uses 20 constant-temperature segments, with boundaries uniformly spaced in f^n. Cantera 3.2.0 integrates the chemistry with relative and absolute tolerances of 10^{-8} and 10^{-15}, respectively. The converted AramcoMech 3.0 mechanism contains 581 species and 3037 reactions.

2.3 Surrogate construction and checks

The training set contains 512 operating conditions across the full domain and 128 additional high-severity conditions, sampled using Latin hypercubes. The high-severity conditions cover 900–1000 °C, 0.2–1.5 s, and steam/ethane ratios of 0–0.40, with the same pressure and heating-exponent ranges. A separate set of 128 conditions across the full domain is excluded from training and used to test the predictions. GP regression[20] uses the inputs $[T_{\text{out}}, \log_{10}\tau, S/H, P, n]$, where S/H is the steam/ethane mass ratio and P is pressure. Inputs are normalized using the observed training extrema. Conversion and molar ethylene selectivity use logit transforms, and species yields and enthalpy rise use logarithmic transforms. An anisotropic radial-basis-function kernel is shared across outputs. Kernel selection and uncertainty calibration use an internal fitting/tuning split.

The selected kernel is then fitted to all 640 training conditions. SI Section S1.3 gives the fitting procedure.

The surrogate predicts ethane conversion, ethylene selectivity, seven mass yields (ethylene, methane, hydrogen, acetylene, ethane, C_3, and C_{4+}), and enthalpy rise. The C_3 and C_{4+} groups sum species with three and at least four carbon atoms, respectively. The surrogate is evaluated against three criteria: $R^2 \geq 0.90$, root-mean-square error divided by the observed holdout range below 0.08, and at least 80% empirical coverage by the nominal 95% prediction intervals for each output. These checks measure how closely the surrogate reproduces the kinetic calculations.

Raw Cantera calculations are checked for carbon, hydrogen, oxygen, and mass closure. A separate five-condition comparison of 20 and 40 segments tests temporal discretization. The GP predictions are checked against the holdout calculations, and the material and energy balances are checked against the raw Cantera outputs.

2.4 Conversion to a product basis

The coil processes fresh and recycled ethane together. Lower ethylene yield therefore increases the coil feed needed per unit product, while lower single-pass conversion increases the unreacted ethane returned to the coil. Let X be ethane conversion and Y_E the reactor ethylene mass yield per unit mass of ethane entering the coil. The downstream acetylene converter adds ethylene and ethane yields ΔY_E and ΔY_A on the same basis. Complete ethane recovery gives the following flows per unit mass of final ethylene:

$$\begin{aligned} Y_{E,\text{final}} &= Y_E + \Delta Y_E \\ m_{\text{coil}} &= \frac{1}{Y_{E,\text{final}}} \\ m_{\text{fresh}} &= \max(0, X - \Delta Y_A) m_{\text{coil}} \\ m_{\text{recycle}} &= m_{\text{coil}} - m_{\text{fresh}} \end{aligned} \tag{2}$$

Species yields after acetylene conversion are divided by $Y_{E,\text{final}}$ to obtain the same basis. Reactor ethylene selectivity is derived from Y_E and X, and the independently predicted selectivity is retained to check consistency. The empirical route has no acetylene yield and uses $\Delta Y_E = \Delta Y_A = 0$.

The total reactor duty includes feed preheating and cracking: a constant-heat-capacity calculation covers 150–650 °C, and Cantera supplies the enthalpy rise from 650 °C to the outlet. An assumed transfer-line-exchanger (TLE) recovery of 70% of the calculated sensible heat is valued at the specified export-steam price. Dilution-steam generation and quench condensation add a separate charge. These assumptions account for heat purchase and recovery without solving a plant heat-exchanger network.

2.5 Process and economic calculation

Water is removed before four-stage compression of the dry reactor gas from a fixed 1.5 bar suction basis to 32 bar. Acetylene conversion after compression targets 5 mol ppm acetylene in the C_2 fraction, with 90% selectivity to ethylene and 10% to ethane. Its hydrogen consumption reduces the fuel gas available for heating or export. The cold section removes light gases, and the C_2 splitter separates ethylene from ethane returned to the coil. Refrigeration work depends on gas flow, light-gas fraction, and the assumed refrigeration efficiency. The splitter load depends on the ethylene/ethane ratio and specified ethylene purity. The cold-box recovery target sets its

cooling requirement. Product flows use the complete-recovery basis in Eq. (2). The equations and constants are summarized in the Supporting Information (SI).

Section-specific scaling relates coil flow, compressor work, refrigeration work, and splitter load to a 126.8 million USD bare-module anchor at 610 kt y^{-1}. Total overnight cost is 2.5 times the scaled bare-module cost. Annual capital and operating charges are divided by production corrected with the implemented run-length correlation. The cost is

$$C_E = C_{\text{feed}} + C_{\text{heat}} + C_{\text{process electricity}} + C_{\text{dilution}} + C_{\text{capital}} + C_{\text{O\&M}} - C_{\text{credits}} \tag{3}$$

The reported cost is the sum of feed, utility, annualized capital, and operating charges after credits for TLE heat recovery, surplus fuel-gas energy, and C_3+ coproducts. These credits use the prices in Table 1. All monetary inputs are fixed scenario values in USD without inflation or location adjustment. The capital anchor is inherited from the repository's interpretation of the published conventional-case investment[13]. Section splits, scaling exponents, and utility shortcuts are retained as model assumptions.

Table 1. Reference screening conditions. One variable is changed at a time unless stated otherwise.

Input	Value
Outlet temperature	850 °C
Residence time	0.35 s
Steam/ethane mass ratio	0.35
Reactor pressure	1.5 bar
Heating exponent	1
Nominal ethylene capacity	610 kt y^{-1}
Fired / electric heat-delivery efficiency	0.60 / 0.95
Ethane price	200 USD t^{-1}
Natural-gas price	4 USD GJ^{-1}
Electricity price	7.0 cents/kWh
Grid emission factor	0.36 kg CO_2e kWh^{-1}
Export-steam credit	9.8 USD GJ^{-1}
C_3+ credit	550 USD t^{-1}
Annual capital charge	0.10
Labor	5 million USD y^{-1}
Ethane upstream emission factor	0.50 kg CO_2e kg^{-1}

2.6 Matched heating-mode and carbon comparisons

The heating comparison holds gas-temperature history, composition, process loads, and capital scaling fixed. Fired heating consumes available tail-gas energy before purchasing natural gas. Electric heating purchases electricity and assigns the unused tail gas a credit at the natural-gas energy price. The heat-delivery efficiency changes from 0.60 to 0.95. Thus the comparison asks what changing the energy source costs for the same reactor output. The separate temperature-history study evaluates changes in that output.

The carbon calculation applies emission factors to purchased gas, eventual tail-gas combustion, process and heating electricity, and upstream ethane. TLE steam and C_3+ receive displacement credits. All plotted cases count eventual combustion of the full methane coproduct, including gas

unused by the cracker furnace. The economic credit assigns a value to that surplus energy, while the carbon calculation counts its eventual combustion. SI Section S3 specifies the emission factors and inventory boundary.

2.7 Literature calibration and paired cost uncertainty

We use two published utility demands to calibrate effective process parameters. For Chen et al., reactor conditions are first selected to match residual ethane, and compressor power per unit ethylene production then calibrates a compression multiplier. For Shin et al., fresh-feed demand constrains the reactor conditions, and total combustion energy calibrates fired efficiency. The calibration assumes uniform parameter priors and equal prior weights for the selected operating conditions. Differences between calculated and reported utilities are represented by a Gaussian distribution with an assumed standard deviation of 5%, 10%, or 20% of the reported value. Each parameter integral uses 161 quadrature nodes.

For each assumed discrepancy level, 512 parameter sets are evaluated for both heating modes at the reference reactor condition. Each pair uses the same compression, heat-recovery, and shared capital factors, reactor predictions, and prices. Electric heating efficiency and its furnace-capital factor are varied in separate scenarios. The resulting cost distributions quantify uncertainty from these process parameters while holding the GP predictions fixed. GP prediction uncertainty is evaluated separately against the holdout calculations. SI Section S1.4 defines the priors, weighting, and paired calculations.

3 Results and discussion

3.1 Accuracy of the kinetic surrogate

For the 128 Cantera calculations excluded from training, the GP predicts all ten outputs with root-mean-square errors below 1.62% of each output's observed range. Ethane conversion, ethylene yield, and enthalpy rise have R^2 values of 0.9997, 0.9994, and 0.9993, respectively, with normalized errors of 0.59%, 0.85%, and 0.77%. The largest error occurs for the C_3 lump. **Figure 2** shows representative parity plots, and all output metrics appear in Table S1.

Across all 768 raw kinetic calculations, the largest elemental residual is 5.69×10^{-11} and the largest total-mass residual is 2.00×10^{-15}. For five recorded test cases, doubling the number of temperature segments keeps changes in conversion, selectivity, and yield below 0.005 and changes in enthalpy below 1%. At the reference condition, increasing from 20 to 40 segments changes conversion by 0.088 percentage points and enthalpy by 0.148%.

Process screening depends on ratios of kinetic outputs. Applying Eq. (2) to the holdout set gives a median absolute relative error of 0.665% in coil ethane flow and a maximum of 15.9%. At low ethylene yield, a small yield error can cause a much larger error in the required coil flow. At the reference condition, the process-derived and directly predicted molar selectivities differ by 0.669 percentage points. The process calculations use selectivity derived from yield and conversion.

3.2 Operating conditions, recycle, and production cost

The reference kinetic-surrogate case predicts 38.0% ethane conversion, a yield of 0.329 kg ethylene per kilogram of ethane entering the coil, and 1.88 kg of recycled ethane per kilogram of ethylene. The empirical reference predicts 65.0% conversion and 0.656 kg kg^{-1} recycle (Table 2). Its rate

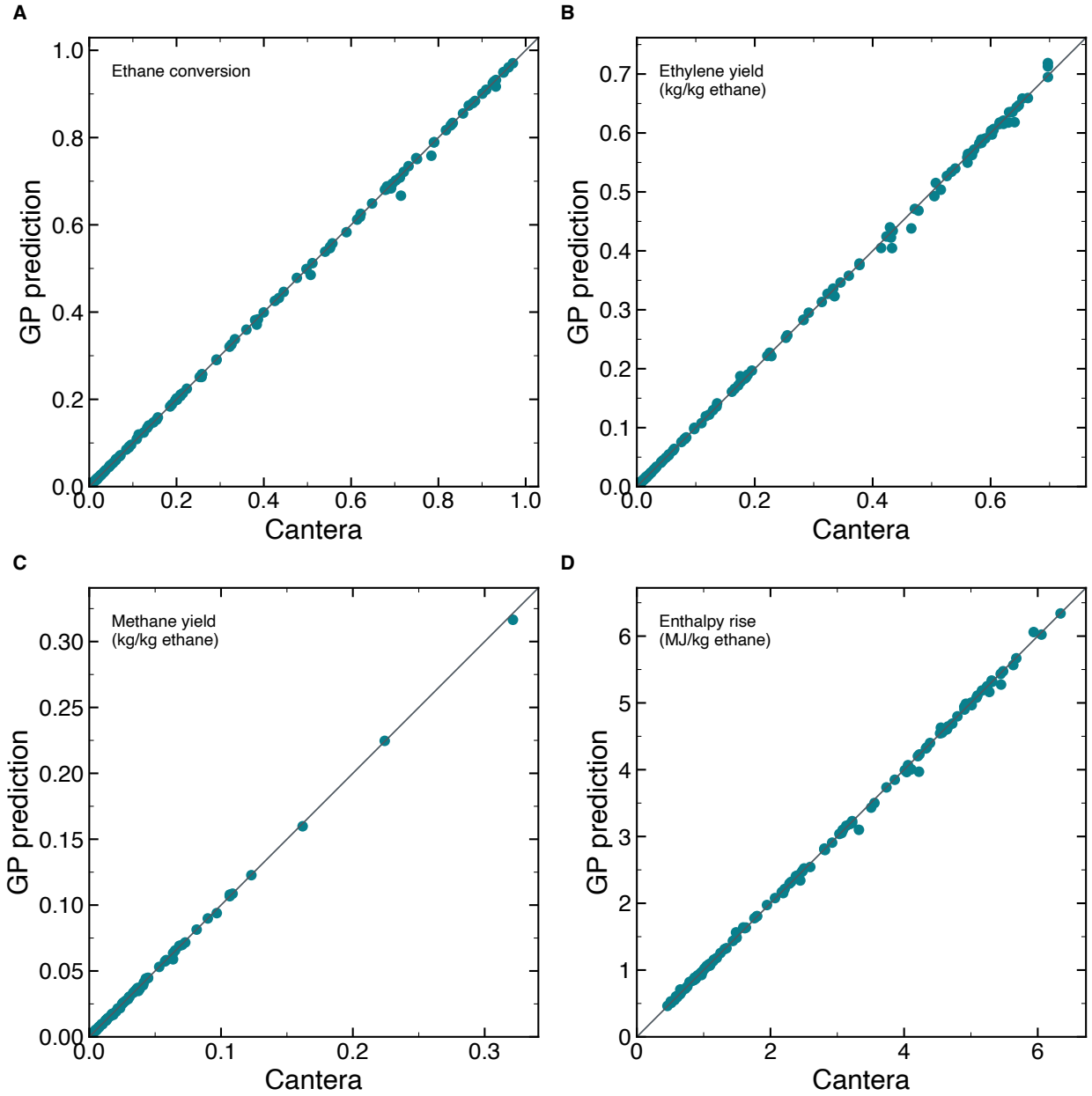


Figure 2. Independent surrogate holdout checks. **(A)** Ethane conversion, **(B)** ethylene mass yield, **(C)** methane mass yield, and **(D)** enthalpy rise for 128 held-out Cantera calculations. Symbols are GP predictions and lines indicate equality. Mass yields and enthalpy use the ethane feed to the coil as their basis.

expression is calibrated at the outlet temperature, whereas the kinetic case follows the 650–850 °C ramp. Both the chemistry description and temperature history therefore differ between these cases.

Table 2. Calculated reference cases. The GP fired and electric cases share the same reactor state. The empirical case evaluates its calibrated rate at the outlet temperature. All cases use Table 1.

Quantity	Empirical fired	GP fired	GP electric
Ethane conversion	0.650	0.380	0.380
Reactor ethylene yield (kg/kg coil ethane)	0.534	0.329	0.329
Fresh ethane (kg/kg ethylene)	1.218	1.150	1.150
Recycled ethane (kg/kg ethylene)	0.656	1.879	1.879
Reactor duty (GJ/t ethylene)	10.87	13.95	13.95
Electricity (MWh/t ethylene)	1.245	1.847	5.927
Overnight cost (million USD)	382	509	509
Cost (USD/t ethylene)	423	503	696
Carbon indicator (t CO_2e/t ethylene)	1.390	1.669	2.407

The GP case requires slightly less fresh ethane because its product selectivity is higher, yet its lower single-pass conversion increases coil flow and recycle. Compression, refrigeration, and heating loads rise on the ethylene-product basis, increasing calculated fired-case cost from 423 to 503 USD t^{-1}. The larger section loads also raise the scaled overnight cost from 382 to 509 million USD.

The heating-history comparison holds outlet temperature at 850 °C and residence time at 0.35 s, with the remaining inputs in Table 1. Increasing the exponent from 0.5 to 4 delays heating and shortens exposure to the hotter part of the 650–850 °C trajectory. Conversion falls from 50.3% to 15.0%, and coil ethane flow rises from 2.37 to 7.66 kg kg^{-1} ethylene. More gas must be reheated, compressed, and separated to produce the same amount of ethylene, raising calculated fired-case cost from 456 to 806 USD t^{-1} (Figure S1).

We next vary outlet temperature and residence time together to map the trade-off between fresh-feed demand and downstream processing loads (**Figure 3**). The scan contains 5,751 conditions over 760–900 °C and 0.20–1.50 s, within the intersection of the prescribed fired/electric operating envelopes and the GP domain. Pressure, dilution, heating exponent, and economic inputs remain at their reference values. Costs in this scan use fired heating.

Maximum GP reactor-outlet ethylene yield occurs at 900 °C and 0.906 s, while minimum calculated cost occurs at 900 °C and 1.50 s. Their costs differ by 5.31 USD t^{-1}, or 1.42% of the maximum-yield point's cost. Both points lie at the upper temperature boundary, and the minimum-cost point also reaches the residence-time ceiling. Relative to the reference case, the minimum-cost point reduces ethane recycle from 1.879 to 0.158 kg kg^{-1} ethylene and cold-box plus splitter electricity from 1,403 to 757 kWh t^{-1}. Fresh ethane demand increases from 1.150 to 1.345 kg kg^{-1} as selectivity declines. Under the stated prices and cost correlations, utility and capital savings together with changed coproduct credits outweigh the additional fresh-feed cost, reducing net cost from 503 to 369 USD t^{-1}.

Passing direct Cantera outputs at R, A, and B through the process model gives costs of 500.6, 375.6, and 367.6 USD t^{-1}, respectively. Direct ethylene yields at A and B are 0.6683 and 0.6565 kg per kilogram of coil-feed ethane, preserving the GP-predicted ordering: longer residence at B lowers ethylene yield while reducing total cost. Increasing the temperature-history discretization from 20 to 40 segments at B changes cost by 0.10 USD t^{-1}.

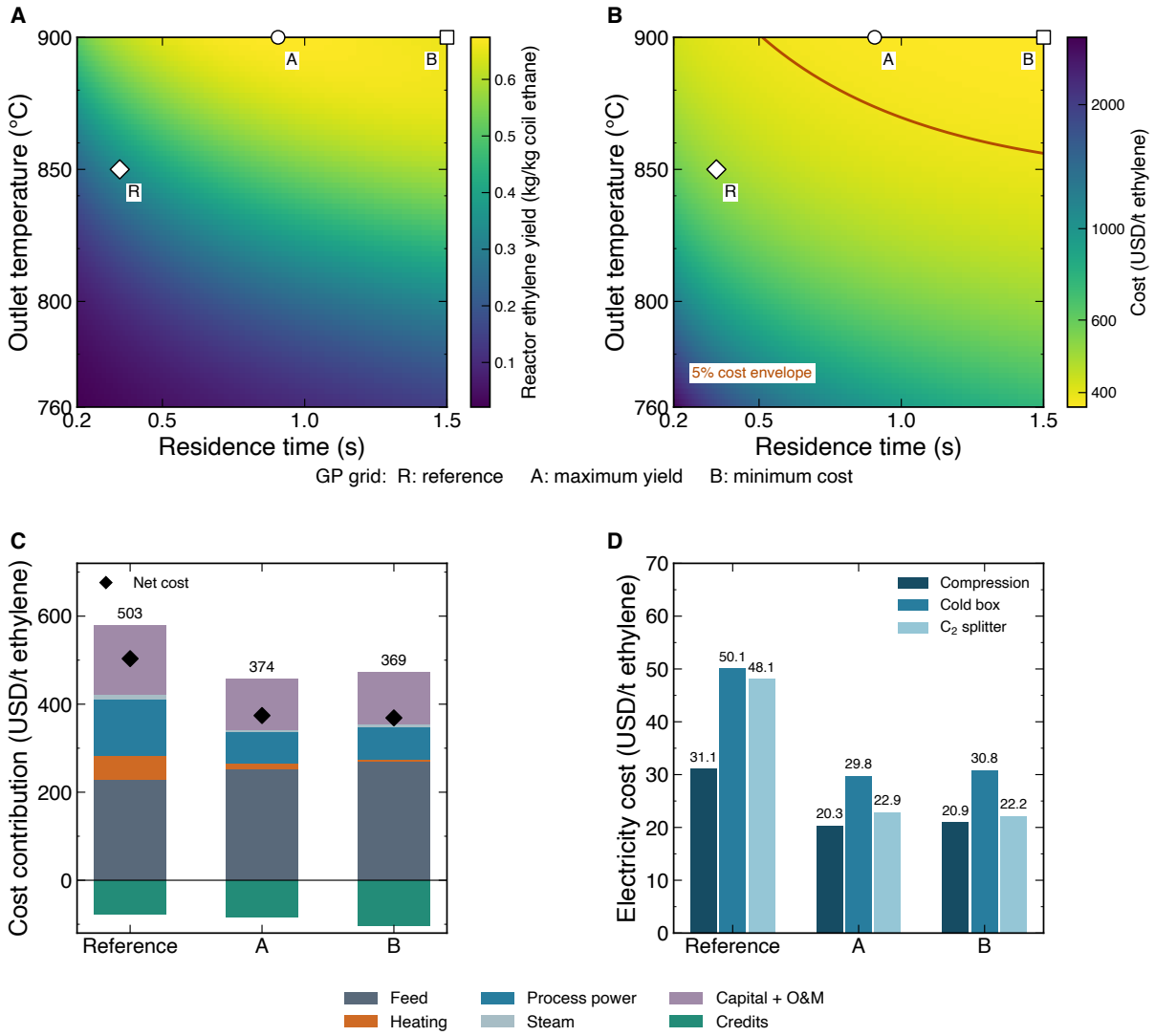


Figure 3. Kinetic and process consequences of operating-condition selection. (A) Reactor-outlet ethylene mass yield and **(B)** calculated fired-case cost over 71 outlet temperatures and 81 logarithmically spaced residence times. Cost uses a logarithmic color scale, and the orange contour marks 5% above the grid minimum. R denotes the reference case (850 °C, 0.35 s), A the maximum-reactor-yield GP grid point (900 °C, 0.906 s), and B the minimum-cost GP grid point (900 °C, 1.50 s). **(C)** Cost contributions at R, A, and B. Black diamonds and numeric labels show net cost after credits. **(D)** Compression, cold-box, and C_2-splitter electricity costs at 7.0 cents/kWh. Blue shades resolve the process-power contribution in (C). Pressure is 1.5 bar, steam/ethane is 0.35 kg kg^{-1}, and the imposed heating exponent is 1. Remaining assumptions follow Table 1.

3.3 Energy-price and carbon crossovers

At the reference reactor condition, electric heating uses 4.080 MWh more electricity per tonne of ethylene than fired heating. At 7.0 cents/kWh, this substitution increases production cost by 193 USD/t ethylene. Feedstock, process electricity, capital, and other shared charges are unchanged, so the cost difference comes from supplying reactor heat.

Let q be the delivered reactor duty in GJ t^{-1} ethylene, p_e the electricity price in USD/kWh, and p_g the gas price in USD/GJ. The equations use USD/kWh, and multiplying by 100 gives the cents/kWh reported in the text and figures. The heat-delivery efficiencies are η_e for electric heating and η_f for fired heating. Because purchased and credited fuel gas share the same energy price, the model gives

$$\Delta C_E = q\left(\frac{p_e}{0.0036\eta_e} - \frac{p_g}{\eta_f}\right) \tag{4}$$

The crossover is therefore

$$p_e^* = 0.0036 p_g \frac{\eta_e}{\eta_f} \tag{5}$$

At a natural-gas price of 4 USD/GJ, electric heating becomes cheaper below 2.28 cents/kWh for fired and electric efficiencies of 60% and 95%, respectively (**Figure 4**A). Reactor duty sets the size of the cost difference but cancels from the break-even price because both routes supply the same duty and value fuel-gas energy at the same price.

At a grid factor of 0.36 kg CO_2e kWh^{-1}, the carbon indicator is 1.669 t CO_2e t^{-1} for fired heat and 2.407 for electric heat. Electric heating has lower calculated emissions below a grid intensity of 0.179 kg CO_2e kWh^{-1} under the default tail-gas-combustion and coproduct-credit assumptions (**Figure 4**B).

3.4 Comparison with published process cases

Three published process cases provide comparisons of composition, feed demand, and utility consumption (**Figure 5**). Where operating conditions are incomplete, a reported quantity is used to select compatible reactor states before comparing the remaining outputs. Matching the residual ethane fraction reported by Chen et al.[1] gives ethylene and hydrogen dry fractions spanning their reported values, while methane ranges from 1.08 to 1.71 times the reported fraction. For Shin et al.,[2] operating conditions selected to match fresh-feed demand give 17.20–18.58 GJ t^{-1} combustion heat against 17.2 GJ t^{-1} reported. The corresponding C_3+ production is 8–20% below the published quantity. Figure 5 marks the quantities used to select these states separately from the outputs compared afterward.

The Wang et al.[21] case shows a larger discrepancy in conversion. Using the residence time calculated from the reported geometry and flow rate, the model predicts at most 36.6% conversion over the tested heating-exponent range, compared with 68.5% reported. The attempted conversion fit reaches the lower exponent bound, $n = 0.45$, and recycle demand is 3.37 times the reported value (**Figure 5**C). Passing the published and GP-predicted outlet inventories through the same separation model gives electricity costs of 75.8 and 115.7 USD per tonne of final ethylene, respectively, a 53% increase (**Figure 5**D). The source case uses a different kinetic model, inlet temperature, and feed composition. SI Section S5 records these inputs and the common separation assumptions.

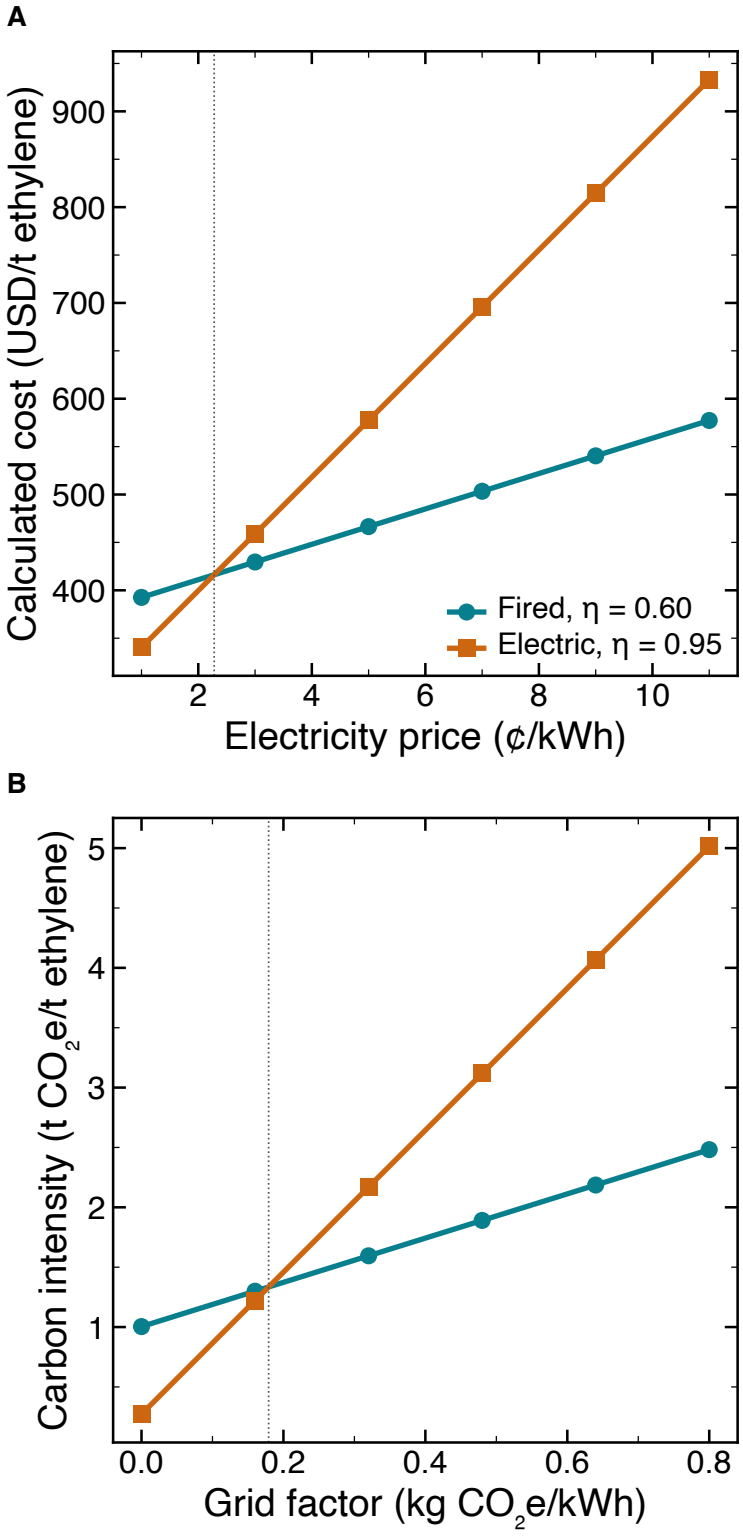


Figure 4. Heating comparisons at the same reactor output. **(A)** Calculated ethylene cost versus electricity price at a gas price of 4 USD GJ^{-1}. **(B)** Inventory-based carbon intensity versus grid emission factor. Fired and electric efficiencies are 0.60 and 0.95, and the reactor follows the reference $n = 1$ profile. Dotted lines mark the price and grid-factor crossovers. Other inputs remain fixed.

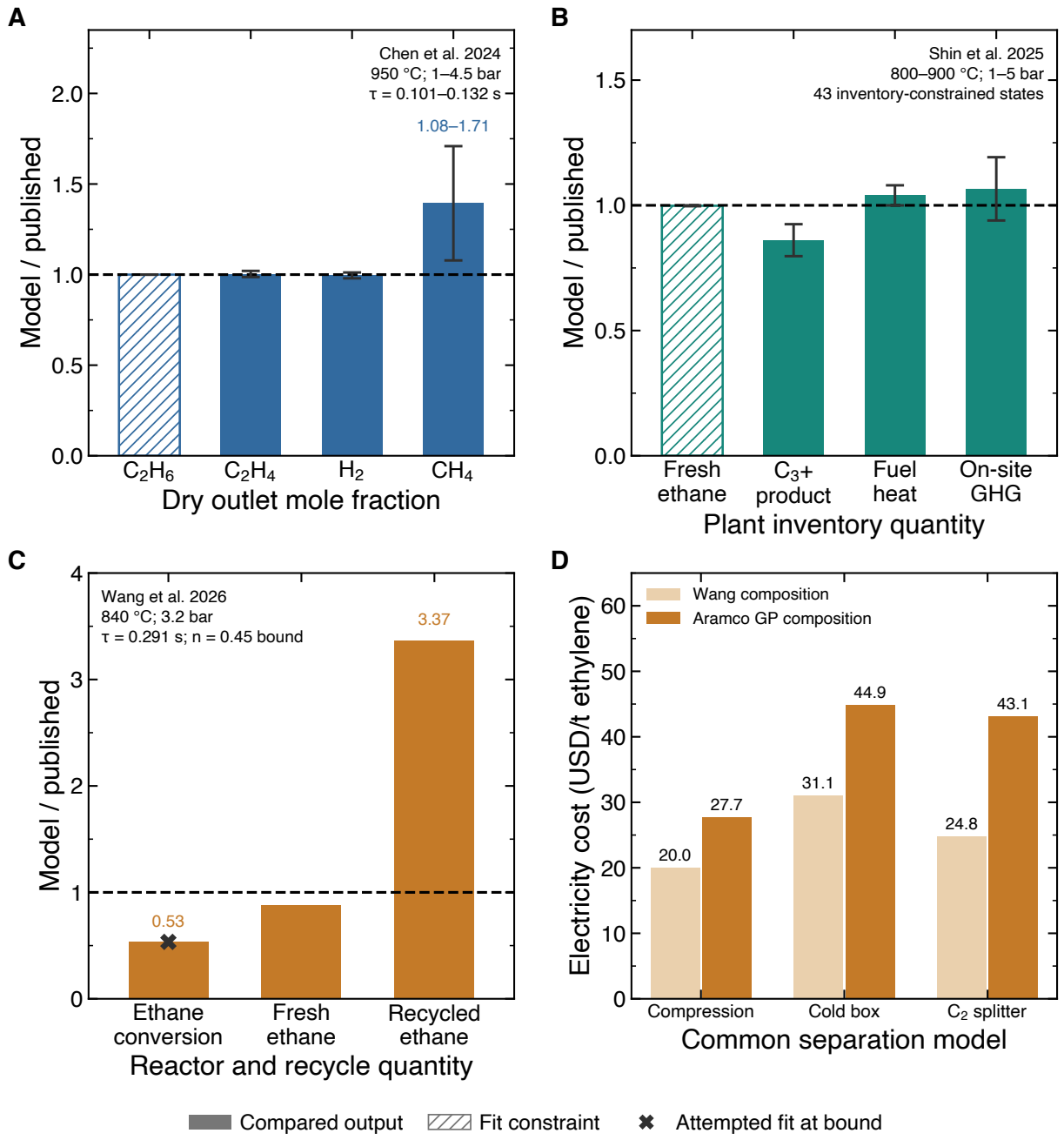


Figure 5. Literature comparisons and separation costs. **(A)** Chen et al.: dry outlet composition with residence time constrained by residual ethane at two pressure interpretations. **(B)** Shin et al.: feed, coproduct, heat, and on-site greenhouse-gas quantities for 43 operating conditions selected to match fresh-feed demand. **(C)** Wang et al.: conversion and fresh and recycled ethane at the geometry-derived residence time. Values in (A–C) are normalized by published quantities. Whiskers span operating envelopes, hatched bars mark quantities used to select the operating conditions, and the cross marks the attempted fit at $n = 0.45$. **(D)** Electricity costs per tonne of final ethylene from Wang's Table 5 and Aramco GP compositions, using identical acetylene-conversion and separation models and 6.0 cents/kWh electricity (SI Section S5). Sources: Chen, Shin, and Wang et al.[1,2,21]

3.5 Uncertainty in the break-even electricity price

Calibrating effective fired efficiency to Shin's combustion demand changes the heating-cost crossover. For the assumed 10% utility discrepancy, calibrated fired efficiency has a median of 63.75%, with a 2.5th–97.5th percentile interval of 52.75–79.00%. With 95% electric heating efficiency and equal furnace capital factors, the median break-even electricity price is 2.16 cents/kWh and its 2.5th–97.5th percentile interval is 1.72–2.57 cents/kWh. Increasing the assumed discrepancy from 5% to 20% widens this interval from 1.93–2.41 to 1.63–2.82 cents/kWh (**Figure 6**A).

At 2.0 cents/kWh, electric heating is cheaper in 93.6%, 78.7%, and 64.1% of sampled parameter sets for the 5%, 10%, and 20% discrepancy cases, respectively (**Figure 6**B). At the reference electricity price of 7.0 cents/kWh, the median electric-heating cost premium is 196–198 USD t^{-1}. Equation (5) explains why the crossover depends on fired efficiency when reactor output and furnace capital factors are the same: shared process costs cancel. Discrepancies in Shin's C_3+ production and Chen's methane fraction and fresh-feed demand persist after utility calibration.

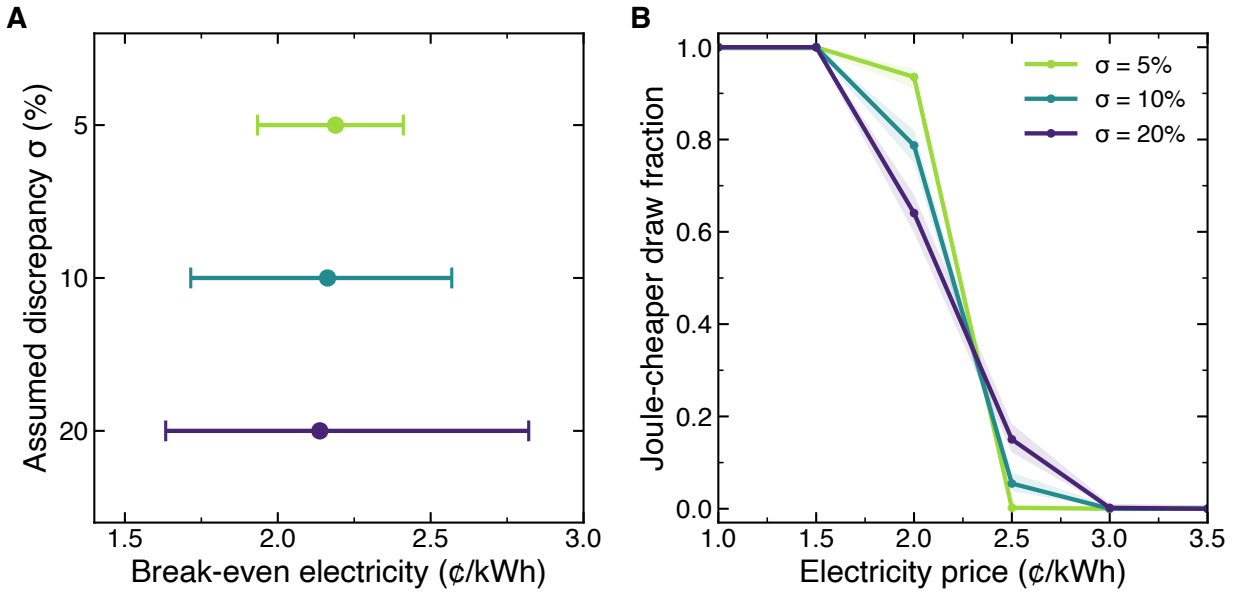


Figure 6. Uncertainty in the break-even electricity price. (A) Median break-even electricity prices and 2.5th–97.5th percentile intervals for assumed utility-discrepancy scales of 5%, 10%, and 20%. **(B)** Fractions of 512 sampled parameter sets with lower electric-heating cost. Shading denotes Wilson 95% Monte Carlo intervals and lines connect evaluated prices. Calculations use the reference reactor condition, 4 USD GJ^{-1} gas, 95% electric heating efficiency, and equal furnace capital factors. Parameter intervals exclude GP posterior uncertainty.

3.6 Editable reactor-to-process calculations

Because gas-temperature history is an input, a heat-transfer model can supply it for a specific reactor configuration. Volume-averaged structured-reactor models, for example, retain temperature-dependent transport and external heat loss at lower computational cost than channel-resolved calculations[22].

Saved scenarios retain the inputs for repeating comparisons. Changing a kinetic mechanism requires new yield and enthalpy maps. Separation correlations can be revised with the kinetic

surrogate held fixed, and prices and grid intensity can be changed with the material and energy inventory held fixed.

Cassady et al. measured species histories during dilute ethane/argon pyrolysis and found temperature-dependent differences among kinetic mechanisms, including underprediction of lower-temperature ethane decomposition by AramcoMech 3.0[23]. The run-length correlation is specified independently of the gas-phase mechanism, so users can revise availability as experimental coking and decoking data become available.

4 Conclusions

Delaying heating at fixed outlet temperature and residence time increases gas circulation and the associated heating and separation loads per tonne of ethylene. In the temperature–residence-time scan, maximum ethylene yield and minimum cost occur at different conditions because lower heating cost and higher coproduct credits outweigh additional feed and processing costs. Literature-case comparisons show how differences in predicted outlet composition propagate to downstream electricity costs. At fixed reactor output and common equipment costs, energy prices and heating efficiencies determine the fired/electric cost crossover.

Code and data availability

The model, training and holdout data, and release records are available at `https://github.com/robin-yk/TEA-EthaneCracking`. The companion manuscript files contain the scenario outputs, literature replay, conditional uncertainty results, reproduction checks, figure scripts, and this TeX source.

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Supporting Information

S1. Reactor equations, surrogate, and numerical checks

S1.1. Constant-pressure species balances

The kinetic calculation follows a closed fluid element with prescribed temperature and constant pressure. Let Y_k denote species mass fraction, W_k molecular weight in kg kmol^{-1}, ρ mixture density, and $\dot{\omega}_k$ net molar production rate in kmol m^{-3} s^{-1}. The integrated species equations are

$$\begin{aligned} \frac{dY_k}{dt} &= \frac{W_k \dot{\omega}_k}{\rho} \\ \dot{\omega}_k &= \sum_r (\nu''_{kr} - \nu'_{kr}) \mathcal{R}_r \\ \rho &= \frac{P\overline{W}}{R_u T} \\ \overline{W}^{-1} &= \sum_k \frac{Y_k}{W_k} \end{aligned} \tag{S1}$$

Here ν'_{kr} and ν''_{kr} are reactant and product stoichiometric coefficients, $\mathcal{R}_r$ is the net rate of reaction r, and R_u is the gas constant on a kmol basis. Cantera evaluates these rates from the released mechanism, including its pressure-dependent and third-body expressions. For an elementary mass-action reaction, the rate has the form $\mathcal{R}_r = k_{f,r} \prod_k c_k^{\nu'_{kr}} - k_{b,r} \prod_k c_k^{\nu''_{kr}}$, where $c_k = \rho Y_k / W_k$. The reaction-specific mechanism definitions determine the actual rate law. The energy equation is disabled because Eq. (1) specifies temperature.

For N_s temperature segments, the implementation sets

$$\begin{aligned} u_i &= \frac{i}{N_s} \\ f_i &= u_i^{1/n} \\ \Delta t_i &= \tau (f_{i+1} - f_i) \\ T_i &= T_{\text{in}} + (T_{\text{out}} - T_{\text{in}}) \left(\frac{f_i + f_{i+1}}{2} \right)^n \end{aligned} \tag{S2}$$

for $i = 0, \ldots, N_s - 1$. Composition passes continuously between segments, and each segment uses constant T_i and P. The release uses $N_s = 20$, with relative and absolute integration tolerances 10^{-8} and 10^{-15}. At the final composition, the thermodynamic state is reset to T_{out} to evaluate outlet enthalpy without an additional reaction interval.

The inlet basis is $n_{A,0} = 1$ kmol ethane, where A denotes ethane and E denotes ethylene. With steam/ethane mass ratio d, inlet ethane mass is $m_{A,0} = W_A n_{A,0}$, steam mass is $d m_{A,0}$, and total

mass is $m_0 = (1+d)m_{A,0}$. Outlet amounts and reactor outputs are

$$\begin{aligned} n_{k,\mathrm{out}} &= \frac{m_0 Y_{k,\mathrm{out}}}{W_k} \\ X &= \frac{\max(0, n_{A,0} - n_{A,\mathrm{out}})}{n_{A,0}} \\ S_E &= \frac{n_{E,\mathrm{out}}}{X n_{A,0}} \\ Y_E &= \frac{m_0 Y_{E,\mathrm{out}}}{m_{A,0}} \\ h_{\mathrm{coil}} &= \frac{m_0(h_{\mathrm{out}} - h_{\mathrm{in}})}{10^6 m_{A,0}} \end{aligned} \tag{S3}$$

The specific enthalpies h_{in} and h_{out} are in J kg^{-1} mixture, so h_{coil} is in MJ kg^{-1} inlet ethane. The code sets $S_E = 0$ when consumed ethane is at most 10^{-12} kmol. Individual species yields use the same mass denominator as Y_E. C_3 and C_{4+} sum masses of species containing exactly three and at least four carbon atoms, respectively. For element a, with b_{ak} atoms per molecule, closure is checked using

$$\begin{aligned} \epsilon_a &= \frac{\sum_k b_{ak} n_{k,\mathrm{out}} - \sum_k b_{ak} n_{k,0}}{\max(\sum_k b_{ak} n_{k,0}, 10^{-30})} \\ \epsilon_m &= \sum_k Y_{k,\mathrm{out}} - 1 \end{aligned} \tag{S4}$$

For an oxygen-free inlet, the oxygen diagnostic is the outlet oxygen amount itself.

S1.2. Input and output transformations

Let $\mathbf{v} = (T_{\mathrm{out}}, \log_{10}\tau, d, P, n)$, with temperature in °C, time in seconds, and pressure in bar. For input coordinate j, normalization uses the observed training extrema a_j and b_j:

$$x_j = \frac{v_j - a_j}{\max(b_j - a_j, 10^{-12})} \tag{S5}$$

The saved surrogate stores these extrema separately from the prescribed sampling-domain limits. For output coordinate o and training row i, the transformations are

$$z_{io} = g_o(y_{io}) = \begin{cases} \log[p_{io}/(1-p_{io})], & p_{io} = \min(1-10^{-7}, \max(10^{-7}, y_{io})) \\ \log[\max(0, y_{io}) + 10^{-10}], & \text{yield or enthalpy} \end{cases} \tag{S6}$$

The first branch applies to conversion and molar selectivity. The transformed training outputs are centered and scaled as $\tilde{z}_{io} = (z_{io} - \mu_o)/s_o$, where μ_o and s_o are the mean and population standard deviation over all 640 training rows. If $s_o < 10^{-12}$, the code uses $s_o = 1$. These normalization constants are retained during the internal fitting/tuning split.

S1.3. Shared covariance, fitting, and prediction

The RBF covariance uses normalized inputs:

$$k(\mathbf{x}, \mathbf{x}') = \exp\left[-\frac{1}{2}\sum_j \left(\frac{x_j - x'_j}{\ell_j}\right)^2\right] \tag{S7}$$

The selected length scales are 0.418, 0.380, 0.532, 0.570, and 0.532 for temperature, logarithmic residence time, dilution, pressure, and heating exponent, respectively. Training and holdout sampling seeds are 42 and 4242. Hyperparameter tuning uses seed 2026. The 39 candidates comprise seven common rescalings of the initial vector $(0.22, 0.20, 0.28, 0.30, 0.28)$ by $(0.45, 0.65, 0.85, 1, 1.2, 1.5, 1.9)$ and 32 independent log-uniform five-coordinate draws between 0.055 and 1.35. All candidate coordinates are restricted to 0.035–2.5.

For a fitting set with covariance matrix $K_{ij} = k(\mathbf{x}_i, \mathbf{x}_j)$, the diagonal regularizer is $\lambda = 2 \times 10^{-6}$. Cholesky factorization and prediction use

$$\begin{aligned} A &= K + \lambda I = LL^\top \\ \alpha_o &= L^{-\top} L^{-1} \tilde{\mathbf{z}}_o \\ \hat{z}_o(\mathbf{x}_*) &= \mu_o + s_o \mathbf{k}_*^\top \alpha_o \\ v_* &= \max(0, 1 - \mathbf{k}_*^\top A^{-1} \mathbf{k}_*) \end{aligned} \tag{S8}$$

Here $\mathbf{k}_*$ contains covariances between the query and fitting points. The point prediction is $\hat{y}_o = g_o^{-1}(\hat{z}_o)$, with inverse maps $g_o^{-1}(z) = 1/(1 + e^{-z})$ for conversion and selectivity, and $g_o^{-1}(z) = \max(0, e^z - 10^{-10})$ for yields and enthalpy. This is the inverse-transformed latent mean. Nonlinear inverse transformation does not in general give the mean in original output units. Outputs share A but have separate coefficient vectors, with no fitted cross-output covariance.

The candidate objective is the mean over the ten outputs of tuning-set RMSE divided by that output's tuning-set range. A seeded permutation divides the 640 training points into 512 fitting and 128 tuning points. Kernel search uses a seeded 256-point subset of the fitting set. After selection, a 512-point fit supplies the tuning residuals and variances used for uncertainty calibration. The deployed coefficients and inverse covariance use all 640 training points. The surrogate also calculates a first-order, calibrated uncertainty estimate

$$\begin{aligned} \sigma_{o,0}(\mathbf{x}_*) &= s_o \sqrt{v_*}\, d_o(\hat{z}_o) \\ d_o(z) &= \begin{cases} p(1-p), & p = (1 + e^{-z})^{-1} \\ e^z, & \text{log-transformed output} \end{cases} \\ c_o &= \mathrm{clip}_{[0.25,20]} \left[\sqrt{\frac{1}{128} \sum_{i \in \mathcal{T}} \left(\frac{\hat{y}_{io} - y_{io}}{\max(\sigma_{io,0}, 10^{-14})} \right)^2} \right] \\ \sigma_o &= c_o \sigma_{o,0} \end{aligned} \tag{S9}$$

The set $\mathcal{T}$ is the 128-point internal tuning subset, and its residuals and uncalibrated uncertainties use the 512-point fit. The nominal interval is $\hat{y}_o \pm 1.96\sigma_o$. The variance excludes an additional observation-noise term. Process calculations use GP point predictions, and Section S1.4 propagates uncertainty in effective process parameters.

For $N_h = 128$ holdout points, the reported statistics are

$$\begin{aligned} \mathrm{RMSE}_o &= \sqrt{N_h^{-1} \sum_i (\hat{y}_{io} - y_{io})^2} \\ \mathrm{NRMSE}_o &= \frac{\mathrm{RMSE}_o}{\max_i y_{io} - \min_i y_{io}} \\ R_o^2 &= 1 - \frac{\sum_i (\hat{y}_{io} - y_{io})^2}{\sum_i (y_{io} - \overline{y}_o)^2} \\ \mathrm{coverage}_o &= N_h^{-1} \sum_i \mathbf{1}\{|\hat{y}_{io} - y_{io}| \le 1.96\sigma_{io}\} \end{aligned} \tag{S10}$$

Here $\overline{y}_o$ is the holdout mean and **1** is the indicator function.

Table S1. Reproduced independent holdout performance. RMSE is normalized by the observed range of each output. Coverage is the fraction of 128 points inside the nominal 95% interval.

Output	R^2	RMSE/range (%)	Coverage (%)
Ethane conversion	0.9997	0.59	96.9
Ethylene molar selectivity	0.9993	0.39	93.0
Ethylene mass yield	0.9994	0.85	92.2
Methane mass yield	0.9996	0.26	92.2
Hydrogen mass yield	0.9995	0.76	90.6
Acetylene mass yield	0.9956	1.24	95.3
Ethane mass yield	0.9986	1.27	95.3
C_3 mass yield	0.9971	1.62	83.6
C_{4+} mass yield	0.9962	0.98	86.7
Enthalpy rise	0.9993	0.77	86.7

The raw-data maximum residuals are 1.28×10^{-11} for carbon, 5.69×10^{-11} for hydrogen, 1.26×10^{-11} for oxygen, and 2.00×10^{-15} for mass. The segment-test acceptance thresholds are 0.005 absolute error for conversion, selectivity, and ethylene mass yield, and 1% relative error for enthalpy.

With the final transforms and kernel held fixed, nested farthest-point training subsets of 64, 128, 256, 512, and 640 points give mean range-normalized errors of 6.19%, 3.63%, 2.40%, 1.13%, and 0.87%, respectively, on the same 128-point holdout set. This recorded subset test measures the effect of training density within the released domain.

The GP output transformations enforce positive yields and bounded conversion and selectivity. They leave cross-output elemental closure unconstrained. For example, the holdout ethane-yield prediction reaches 1.023 kg kg^{-1}, while its direct kinetic value stays below unity. The process recycle calculation uses conversion and ethane formed by acetylene hydrogenation, as shown in Eq. (2). The GP ethane-yield coordinate enters the C_2 inventory used to set the converter target. The retained scenario files include the root-model diagnostic flags. The reference conversion warning uses the legacy empirical range of 0.45–0.85 even when the GP inputs are inside the kinetic training domain. This warning is separate from the GP domain check.

Kim et al. incorporate differential-equation residual penalties into GP marginal-likelihood hyperparameter estimation[24]. Adding joint yield-balance residuals to the present tuning objective is a candidate extension. The released model selects hyperparameters using prediction error alone.

S1.4. Conditional utility calibration and paired uncertainty

Chen's reported compressor power, 29.4 MW, divided by ethylene production, 79.4 t h^{-1}, sets the compression observation. Residence time first matches the reported dry ethane fraction at either 1 or 4.5 bar, and the two pressure interpretations receive equal prior weight. Shin's 17.2 GJ t^{-1} total combustion energy sets the heat observation. The corresponding operating-condition scan covers 800–900 °C in 5 °C steps and 1–5 bar in 0.5 bar steps, with residence time in 0.02–1.5 s fitted to fresh-feed demand. States within 0.5% of that demand receive equal prior weight. The reported residual-ethane fraction and fresh-feed demand are treated as fixed inputs when selecting these conditions.

Let z_j denote a reconstructed state, d_o the utility observation, and $F_o(\theta_i, z_j)$ the corresponding

model output at parameter node θ_i. A uniform prior and Gaussian aggregate mismatch give normalized quadrature weights

$$\begin{aligned} \widetilde{w}_{ij} &= a_i \exp\left[-\frac{(F_o(\theta_i, z_j) - d_o)^2}{2\sigma_o^2}\right] \\ w_{ij} &= \frac{\widetilde{w}_{ij}}{\sum_{i',j'} \widetilde{w}_{i'j'}} \\ \sigma_o &= \delta d_o \\ \delta &\in \{0.05, 0.10, 0.20\} \end{aligned} \tag{S11}$$

Each parameter grid has 161 evenly spaced nodes, with trapezoidal weights $a_i = 1/2$ at its endpoints and 1 elsewhere. Fired efficiency has prior $\eta_f \sim U(0.45, 0.85)$, and the compression multiplier has prior $k_c \sim U(0.5, 2.0)$. Their outputs are $F_{\text{heat}} = q_{\text{demand}}(z)/\eta_f$ and $F_{\text{comp}} = k_c w_c(z)$ in units matching each observation. The two studies are treated as conditionally independent. These parameters absorb differences in equipment and process boundaries. The mismatch scales and prior bounds are analyst assumptions. Other species and emission outputs from the same studies serve as held-out checks. Their reported intervals contain parameter variation without added mismatch noise. Financial quantities and Wang's different reactor boundary do not enter the likelihood.

Each scenario draws 512 parameter pairs using seed 20260923. The fitted η_f and k_c are sampled from their marginal quadrature weights. Heat-recovery fraction $r_h \sim U(0.5, 0.9)$ and shared capital multiplier $k_I \sim U(0.7, 1.5)$ are sampled independently and retain their priors. Electric efficiency is fixed at 0.90, 0.95, or 0.98, and its furnace-only capital multiplier at 0.7, 1.0, or 1.3. Together with three mismatch scales, these choices define 27 scenarios. Both routes receive the same draw of k_c, r_h, k_I and the same reference chemistry and prices. The calculation holds GP outputs at their point predictions.

For parameter set b, cost varies linearly with electricity price p_e. Evaluations at 0 and 10 cents/kWh (0 and 0.1 USD/kWh in the equations) determine

$$\begin{aligned} A_b &= C_{e,b}(0) - C_{f,b}(0) \\ B_b &= \frac{C_{e,b}(0.1) - C_{f,b}(0.1) - A_b}{0.1} \\ \Delta C_b(p_e) &= A_b + B_b p_e \\ p^*_{e,b} &= -A_b/B_b \end{aligned} \tag{S12}$$

Here e and f denote electric and fired routes. Quantiles at 2.5%, 50%, and 97.5% summarize absolute costs, paired differences, and crossing prices. The fraction with $\Delta C_b < 0$ estimates the conditional frequency of lower electric cost, with a Wilson interval for Monte Carlo counting uncertainty. Common process charges cancel in the difference when furnace factors match, and the crossing then reduces to Eq. (5). A route-specific furnace factor retains a differential capital term.

At 10% assumed mismatch, the fitted fired-efficiency interval is 52.75–79.00% and the compression-multiplier interval is 0.988–1.475. Shin's C_3+ quantity, 0.1376 kg kg^{-1} ethylene, remains above the conditional interval of 0.1114–0.1270. Chen's methane fraction, fresh-feed demand, and reactor heat also remain outside their parameter intervals.

S2. Implemented process correlations

S2.1. Material and heat balances

All process masses below are kg per kg final ethylene, and all process duties and work are kJ per kg final ethylene. Let y_k^{coil} be a GP species yield on the inlet-ethane basis and W_k its molecular weight in g mol^{-1}. The acetylene converter operates on this basis before the plant flows are scaled. With A denoting ethane and E ethylene, its molar balances are

$$\begin{aligned}
\nu_k &= 1000 y_k^{\mathrm{coil}}/W_k \\
\nu_{\mathrm{C_2}} &= \nu_E + \nu_A + \nu_{\mathrm{C_2H_2}} \\
\nu_{\mathrm{res}} &= \min(\nu_{\mathrm{C_2H_2}}, 5\times10^{-6}\nu_{\mathrm{C_2}}) \\
\xi &= \nu_{\mathrm{C_2H_2}} - \nu_{\mathrm{res}} \\
\Delta Y_E &= 0.90\xi W_E/1000 \\
\Delta Y_A &= 0.10\xi W_A/1000 \\
Y_{\mathrm{H_2,need}} &= 1.10\xi W_{\mathrm{H_2}}/1000
\end{aligned} \tag{S13}$$

The quantities ν_k and ξ are mol per kg coil-feed ethane. The 5 ppm target and 90% ethylene selectivity are fixed assumptions. Available process hydrogen is consumed first. Any shortfall is recorded as external hydrogen, so hydrogen availability does not reduce the imposed acetylene conversion. External hydrogen purchase and converter heat release are omitted from the cost and heat balances, respectively. Converter capital is included in the existing recovery-section anchor. Post-converter masses are

$$\begin{aligned}
m_{\mathrm{H_2}} &= \max(0, y_{\mathrm{H_2}}^{\mathrm{coil}} - Y_{\mathrm{H_2,need}}) m_{\mathrm{coil}} \\
m_{\mathrm{C_2H_2}} &= (\nu_{\mathrm{res}} W_{\mathrm{C_2H_2}}/1000) m_{\mathrm{coil}} \\
m_k &= y_k^{\mathrm{coil}} m_{\mathrm{coil}} \quad (k = \mathrm{CH_4}, \mathrm{C_3}, \mathrm{C_{4+}}) \\
m_s &= d\, m_{\mathrm{coil}} \\
S_{E,\mathrm{process}} &= \frac{Y_E W_A}{\max(X, 10^{-12}) W_E} \\
S_{E,\mathrm{mass}} &= \frac{Y_E}{\max(X, 10^{-12})}
\end{aligned} \tag{S14}$$

Here m_s is dilution steam and d is its feed mass ratio to ethane. Solid-coke mass m_C is zero in this route. A difference exceeding 0.01 between derived and independently predicted molar selectivity triggers a diagnostic. The 0.995 separation-recovery specification enters the cold-box correlation below. The product-basis mass balance has no additional recovery-loss multiplier.

Upstream preheat uses heat capacities of 3.0 and 2.3 kJ kg^{-1} K^{-1} for ethane and steam. Total sensible heat is computed from 150 °C to the outlet temperature. The TLE credit is 0.70 times this sensible duty. Dilution-steam generation uses 2100 kJ kg^{-1} and is charged at $0.45 p_{\mathrm{steam}}$ +

0.40 USD GJ^{-1}, where the latter term represents cooling-water rejection.

$$\begin{aligned} C_p^* &= 3.0 m_{\text{coil}} + 2.3 m_s \\ q_{\text{sens}} &= C_p^* (T_{\text{out}} - 150) \\ q_{\text{demand}} &= C_p^* (650 - 150) + 1000 m_{\text{coil}} h_{\text{coil}} \\ q_{\text{rxn}} &= q_{\text{demand}} - q_{\text{sens}} \\ q_{\text{input}} &= q_{\text{demand}} / \eta_h \\ q_{\text{TLE}} &= 0.70 q_{\text{sens}} \\ q_{\text{DSG}} &= 2100 m_s \end{aligned} \tag{S15}$$

Temperature differences use °C or K interchangeably, and η_h is heat-delivery efficiency. Disabling heat recovery sets $q_{\text{TLE}} = 0$. The quantity q_{rxn} is the residual after the constant-heat-capacity sensible partition. The GP total duty uses the mechanism enthalpy directly.

Tail-gas energy and heating-mode balances are

$$\begin{aligned} q_f &= 1000(50 m_{\text{CH}_4} + 120 m_{\text{H}_2}) \\ (q_{\text{buy}}, q_{\text{sell}}, w_h)_{\text{fired}} &= (\max(0, q_{\text{input}} - q_f), \max(0, q_f - q_{\text{input}}), 0) \\ (q_{\text{buy}}, q_{\text{sell}}, w_h)_{\text{electric}} &= (0, q_f, q_{\text{input}}) \end{aligned} \tag{S16}$$

The methane and hydrogen lower heating values are 50 and 120 MJ kg^{-1}. The symbols q_{buy} and q_{sell} denote purchased and exported fuel energy, and w_h denotes electric heating input.

S2.2. Compression and refrigeration

Compression uses the dry gas before acetylene conversion. On the final-product basis, its ethylene and ethane masses are $m_E^{\text{raw}} = \max(0, 1 - \Delta Y_E m_{\text{coil}})$ and $m_A^{\text{raw}} = \max(0, m_{\text{recycle}} - \Delta Y_A m_{\text{coil}})$. Raw hydrogen and acetylene masses are $m_k^{\text{raw}} = y_k^{\text{coil}} m_{\text{coil}}$. Methane, C_3, and C_{4+} masses remain unchanged. The total molar amount is $\mathcal{N}_g^{\text{raw}} = \sum_k 1000 m_k^{\text{raw}} / W_k$. The molecular weights of ethylene, ethane, methane, hydrogen, acetylene, C_3, and C_{4+} are 28.05, 30.07, 16.04, 2.016, 26.038, 42.08, and 56.11 g mol^{-1}, respectively, and the last two represent lumped products.

Compression uses four equal pressure-ratio stages with complete intercooling to 313 K, heat-capacity ratio $\gamma = 1.32$, and efficiency $\eta_c = 0.75$:

$$\begin{aligned} r_s &= \left(\frac{32}{1.5} \right)^{1/4} \\ w_c &= \frac{4\gamma R (313) \mathcal{N}_g^{\text{raw}}}{1000(\gamma - 1)\eta_c} \left(r_s^{(\gamma-1)/\gamma} - 1 \right) \end{aligned} \tag{S17}$$

Here $R = 8.314$ J mol^{-1} K^{-1}. The suction pressure remains 1.5 bar even if the reactor-pressure input changes.

The cold box uses the post-converter amounts $\mathcal{N}_k = 1000 m_k / W_k$, with $m_E = 1$ and $m_A = m_{\text{recycle}}$. The total molar amount is $\mathcal{N}_g = \sum_k \mathcal{N}_k$ and the total mass is $m_g = \sum_k m_k$. The light-gas mole fraction is $\phi = (\mathcal{N}_{\text{CH}_4} + \mathcal{N}_{\text{H}_2}) / \mathcal{N}_g$. With specified recovery $r_E = 0.995$, the cold-end

temperature t_c in °C and cold-box duty are

$$\begin{aligned} t_c &= -100 - 40\log_{10}\left(\frac{0.01}{1-r_E}\right) - 160(\phi - 0.40) \\ q_c &= 2.2 m_g (30 - t_c) + 350(1 + m_{\text{recycle}} + m_{\text{C}_2\text{H}_2} + m_{\text{C}_3} + m_{\text{C}_{4+}}) \\ w_r &= q_c \sum_{j=1}^{3} \frac{a_j}{\max[0.05, \eta_2 T_j/(T_a - T_j)]} \end{aligned} \tag{S18}$$

The assumed sensible and latent coefficients are 2.2 kJ kg^{-1} K^{-1} and 350 kJ kg^{-1}, respectively. The duty fractions are $\mathbf{a} = (0.55, 0.30, 0.15)$ at $\mathbf{T} = (233.15, 173.15, t_c{+}273.15)$ K. Ambient temperature is $T_a = 303$ K and second-law efficiency is $\eta_2 = 0.35$. The denominator is a Carnot coefficient of performance multiplied by η_2, with the implemented numerical floor of 0.05. The temperature and duty relations are screening correlations.

S2.3. C_2 splitter shortcut

For discharge pressure $P_d = 32$ bar, relative volatility $\alpha_v = 1.85 - 0.018 P_d$ and distillate ethylene mole fraction $x_D = 0.999$ give

$$\begin{aligned} N_{\min} &= \frac{\log[(x_D/(1-x_D))(0.99/0.01)]}{\log \alpha_v} \\ N &= \text{round}(2.1 N_{\min}) \\ z_F &= \min\left(0.995, \frac{\mathcal{N}_E}{\mathcal{N}_E + \mathcal{N}_A}\right) \end{aligned} \tag{S19}$$

The Fenske expression assumes a 0.01 ethylene mole fraction in the bottoms. The capped binary-feed mole fraction z_F enters the implemented reflux shortcut:

$$\begin{aligned} R_{\min} &= \max\left[0.5, \frac{x_D/z_F - \alpha_v(1-x_D)/(1-z_F)}{\alpha_v - 1}\right] \\ g_N &= \frac{N - N_{\min}}{N+1} \\ R_s &= \max\left[1.15 R_{\min}, 1.05\{R_{\min} + 0.9(1 - \sqrt{g_N})\}\right] \\ w_s &= \frac{280(R_s+1)}{\max[0.05, \eta_2(243)/(T_a - 243)]} \end{aligned} \tag{S20}$$

Here R_s is the operating reflux ratio and w_s is splitter condenser refrigeration work. The 280 kJ kg^{-1} latent-duty coefficient and 243 K condenser temperature are assumed. This algebraic shortcut supplies the utility and capital loads used in the comparison.

S2.4. Capital, availability, and unit cost

The scaled bare-module cost, in million USD, is

$$\begin{aligned} B = 126.8 \left(\frac{K}{610}\right)^{0.65} \Bigg[& 0.30 \left(\frac{m_{\text{coil}}}{1.809}\right)^{0.65} + 0.25 \left(\frac{w_c}{1134}\right)^{0.70} \\ & + 0.25 \left(\frac{w_r}{2098}\right)^{0.70} + 0.20 \left(\frac{R_s N}{195}\right)^{0.60} \Bigg] \end{aligned} \tag{S21}$$

Here K is nominal capacity in kt y^{-1}, w_c and w_r are compression and refrigeration work in kJ kg^{-1} ethylene, R_s is reflux ratio, and N is the actual stage count. Overnight cost is $2.5B$. Annual nonlabor operation and maintenance are 7.5% of overnight cost.

The run-length correlation, in days, is

$$d_{\mathrm{run}} = 60\exp[-0.025(T_{\mathrm{out}} - 850)]\left(\frac{S/H}{0.35}\right)^{0.6}\left(\frac{0.35}{\tau}\right)^{0.3} \tag{S22}$$

The on-stream factor is $d_{\mathrm{run}}/(d_{\mathrm{run}}+1.5)$, with an assumed 1.5-day shutdown between runs. It equals 0.9756 in all three reference cases and throughout the heating-exponent sweep. This availability correlation is an assumed process input. The GP chemistry uses gas-phase reactions and carries zero solid-coke yield in the process interface.

Let F_{on} be this on-stream factor, $M = 1000KF_{\mathrm{on}}$ annual production in t y^{-1}, $I = 2.5B \times 10^6$ overnight cost in USD, f_c annual capital charge fraction, and L_a annual labor cost in USD y^{-1}. With electricity price p_e in USD kWh^{-1}, gas and steam prices p_g, p_s in USD GJ^{-1}, and ethane and C$_3$+ prices p_A, p_L in USD t^{-1}, all cost terms in USD t^{-1} ethylene are

$$\begin{aligned}
C_{\mathrm{feed}} &= m_{\mathrm{fresh}}p_A \\
C_{\mathrm{heat}} &= 10^{-3}q_{\mathrm{buy}}p_g + \frac{w_h}{3.6}p_e \\
C_{\mathrm{process\ electricity}} &= \frac{w_c + w_r + w_s}{3.6}p_e \\
C_{\mathrm{dilution}} &= 10^{-3}q_{\mathrm{DSG}}(0.45p_s + 0.40) \\
C_{\mathrm{capital}} &= \frac{f_c I}{M} \\
C_{\mathrm{O\&M}} &= \frac{0.075I + L_a}{M} \\
C_{\mathrm{credits}} &= 10^{-3}(q_{\mathrm{TLE}}p_s + q_{\mathrm{sell}}p_g) + (m_{\mathrm{C_3}} + m_{\mathrm{C_{4+}}})p_L
\end{aligned} \tag{S23}$$

Equation (3) combines these terms. The factor 10^{-3} converts kJ kg^{-1} to GJ t^{-1}, and division by 3.6 converts the same duty to kWh t^{-1}. The displayed feed-cost decomposition is $C_{\mathrm{floor}} = (W_A/W_E)p_A$ and $C_{\mathrm{selectivity\ loss}} = [m_{\mathrm{fresh}} - W_A/W_E]p_A$, and their sum is C_{feed}. Total electricity demand is $E_t = (w_c + w_r + w_s + w_h)/3.6$ kWh t^{-1}, and the reported net-energy indicator is $10^{-3}(q_{\mathrm{input}} - q_{\mathrm{TLE}} + w_c + w_r + w_s)$ GJ t^{-1}.

S2.5. Empirical reference chemistry

The empirical comparison uses first-order conversion with an activation energy of 272.8 kJ mol^{-1} and a pre-exponential factor chosen to give $X = 0.65$ at 850 °C and 0.35 s. Its quadratic selectivity relation passes through $(X, S) = (0.50, 0.93)$, $(0.65, 0.88)$, and $(0.80, 0.82)$, with added dilution and residence-time corrections. These coordinates are assumed anchors in the source implementation. They define the empirical reference shown in Table 2.

$$\begin{aligned}
A_0 &= \frac{-\log(1-0.65)}{0.35}\exp\left[\frac{272800}{8.314(850+273.15)}\right] \\
X &= \max\{10^{-4}, 1 - \exp[-A_0 e^{-272800/[8.314(T_{\mathrm{out}}+273.15)]}\tau]\} \\
S &= \mathrm{clip}_{[0.55,0.985]}\left[\sum_{i=1}^{3} S_i \prod_{j\neq i}\frac{X - X_j}{X_i - X_j} + 0.06(d - 0.35) + 0.04\log(0.35/\tau)\right]
\end{aligned} \tag{S24}$$

For lost-selectivity fraction $\ell = 1 - S$, the assumed carbon fractions going to methane, coke, and C_3 are

$$\begin{aligned} f_M &= \min(0.80, 0.45 + 0.35\ell/0.30) \\ f_C &= \min(0.25, 0.08 + 0.15\ell/0.30) \\ f_3 &= \max(0.02, 1 - f_M - f_C) \end{aligned}$$

Per mole of consumed ethane, product amounts are

$$\begin{aligned} b_M &= 2\ell f_M \\ b_C &= 2\ell f_C \\ b_3 &= 2\ell f_3/3 \\ b_H &= 3 - 2S - 2\ell(2f_M + f_3) \end{aligned} \tag{S25}$$

The required consumed ethane is $\mathcal{N}_A^{\text{consumed}} = 1000/(SW_E)$ mol per kg ethylene. The resulting masses are

$$\begin{aligned} m_{\text{fresh}} &= \mathcal{N}_A^{\text{consumed}} W_A/1000 \\ m_k &= \mathcal{N}_A^{\text{consumed}} b_k W_k/1000 \\ m_{\text{coil}} &= m_{\text{fresh}}/X \end{aligned}$$

The empirical C_3 lump uses propylene stoichiometry and $W_C = 12.011$ g mol^{-1} for coke. Acetylene and C_{4+} are zero. The empirical heat duty is $q_{\text{demand}} = 145\mathcal{N}_A^{\text{consumed}} + q_{\text{sens}}$, using an assumed 145 kJ mol^{-1} reaction heat. Subsequent utility, capital, and cost equations are unchanged.

S3. Carbon accounting for the plotted scenarios

The gas emission factor is 56.1 kg CO_2 GJ^{-1}. Methane combustion uses 44.01/16.04 kg CO_2 per kilogram of methane. Upstream ethane is assigned 0.50 kg CO_2e kg^{-1}. The avoided-steam factor is 56.1/0.90 kg CO_2 GJ^{-1}, and the avoided-C_3+ factor is 0.80 kg CO_2e kg^{-1}. These scenario factors are applied consistently in the plotted comparison. For the selected eventual-combustion fate and system-expansion credits, the carbon indicator in kg CO_2e per tonne ethylene is

$$\begin{aligned} \text{CI} = 10^{-3} q_{\text{buy}}(56.1) + 1000 m_{\text{CH}_4} \frac{44.01}{16.04} + 1000 m_C \frac{44.01}{12.011} + E_t e_g \\ + 1000 m_{\text{fresh}} e_A - 10^{-3} q_{\text{TLE}} \frac{56.1}{0.90} - 1000(m_{\text{C}_3} + m_{\text{C}_{4+}})(0.80) \end{aligned} \tag{S26}$$

Here e_g is grid intensity in kg CO_2e kWh^{-1} and $e_A = 0.50$ kg CO_2e kg^{-1} ethane. The coke term vanishes for GP chemistry. Methane consumed internally and methane exported for eventual combustion sum to the full methane yield in this boundary. At matched chemistry and heat demand, both cost and CI are affine in electricity price and grid intensity, respectively. Writing $\Delta C_E = \Delta C_0 + \Delta E_t p_e$ and $\Delta\text{CI} = \Delta\text{CI}_0 + \Delta E_t e_g$, with electric minus fired differences, gives crossover coordinates $p_e^* = -\Delta C_0/\Delta E_t$ and $e_g^* = -\Delta\text{CI}_0/\Delta E_t$ when $\Delta E_t \neq 0$. The zero-subscript terms collect all terms independent of the varied coordinate.

The inventory omits construction, upstream purchased-gas emissions, and a separate burden for dilution-steam generation.

S4. Reproduction

The implementation corresponding to Sections S1–S3 is organized as follows: `multiscale/cantera_sweep.py`, function `simulate_case`, defines the segmented reactor and output basis. `multiscale/train_gp.py` defines transformations, tuning, fitting, and uncertainty calibration. `multiscale/gp_runtime.js` evaluates full surrogate predictions and uncertainties. The `mechanismGP`, `model`, and `lca` functions in the repository-root `engine.html` implement the process point predictions, utility and economic calculations, and carbon inventory. The serialized `multiscale/surrogate.json` contains the normalization constants, training coordinates, coefficient matrix, inverse covariance, and output calibration factors needed to reproduce released predictions without retraining.

The reference case is evaluated with both GP and empirical chemistry. The ramp, electricity-price, and grid-factor sweeps use GP chemistry. Holdout predictions and raw-data closure are checked against the Cantera calculations. Direct Cantera integrations are repeated at R, A, and B, including a 40-segment refinement at B. The training set contains 640 conditions, and the segment-convergence comparison covers five conditions.

S5. Comparison with published process cases

The repository benchmark inputs and fitting routines were rerun with the current surrogate. The comparisons use the published quantities transcribed in `benchmarks/literature.json`. Wang's Table 5 outlet composition was checked against the source article for the downstream comparison in Figure 5D.

For the Chen et al. case (Green Chemistry, 2024), residence time is constrained using a residual ethane dry mole fraction of 0.151 at 950 °C. The two reported pressure interpretations, 1 and 4.5 bar, give residence times of 0.101–0.132 s. Ethylene and hydrogen dry mole fractions are 0.372–0.385 and 0.399–0.412, against reported values of 0.377 and 0.407. Fresh ethane demand is 1.185–1.219 versus 1.108 kg kg^{-1} ethylene. The model's 650 °C-to-outlet enthalpy demand is 7.60–7.68 GJ t^{-1}, while the reported reactor heat is 5.25 GJ t^{-1}, and their temperature boundaries must be aligned to interpret that difference.

For the Shin et al. case (Green Chemistry, 2025), residence time is constrained by fresh ethane demand over 800–900 °C and 1–5 bar. Forty-three states satisfy the 0.5% feed tolerance, with residence times of 0.666–1.50 s. At the default process parameters, combustion-heat and on-site-emission ranges are 17.20–18.58 GJ t^{-1} and 415–527 kg CO_2e t^{-1}, compared with 17.2 and 442 reported. C_3+ yield is 0.1097–0.1273 versus 0.1376 kg kg^{-1}. The subsequent uncertainty analysis uses Shin's combustion heat and Chen's compressor work to calibrate effective utility parameters.

Wang et al. (Frontiers in Energy Research, 2026) report tube geometry and feed flow giving a mean-state residence time of 0.291 s. At 840 °C and 3.2 bar, the maximum conversion in the tested heating-exponent range is 0.366 at $n = 0.45$, against 0.685 reported. The paper uses a different kinetic model, a 502.77 °C inlet, and a feed containing 0.64 wt% ethylene, whereas the surrogate starts at 650 °C with ethane and steam. The replay retains economic outputs as context because capital, financing, and coproduct boundaries differ among studies. For Figure 5D, Wang's Table 5 P-OUT mass fractions define the raw outlet inventory, including an ethylene mass fraction of 0.3995. Both inventories pass through AC-201 and are normalized by their final ethylene production. The dry inventory includes ethane, ethylene, hydrogen, methane, acetylene, C_3, and C_{4+}, and steam is removed before compression. C_3 combines propylene and propane, and C_{4+} combines the listed C_4 species and benzene. Both inventories use the equations in Section S2, including compression from 1.5 to 32 bar, the same acetylene-conversion assumptions, and common recovery and purity

specifications. Compression, cold-box, and splitter costs are 20.0, 31.1, and 24.8 USD t^{-1} for the published inventory and 27.7, 44.9, and 43.1 USD t^{-1} for the GP inventory. These costs quantify the composition effect within the shared equipment model. The five outlet-species comparisons remain in the accompanying replay data.

S6. Heating-history sensitivity

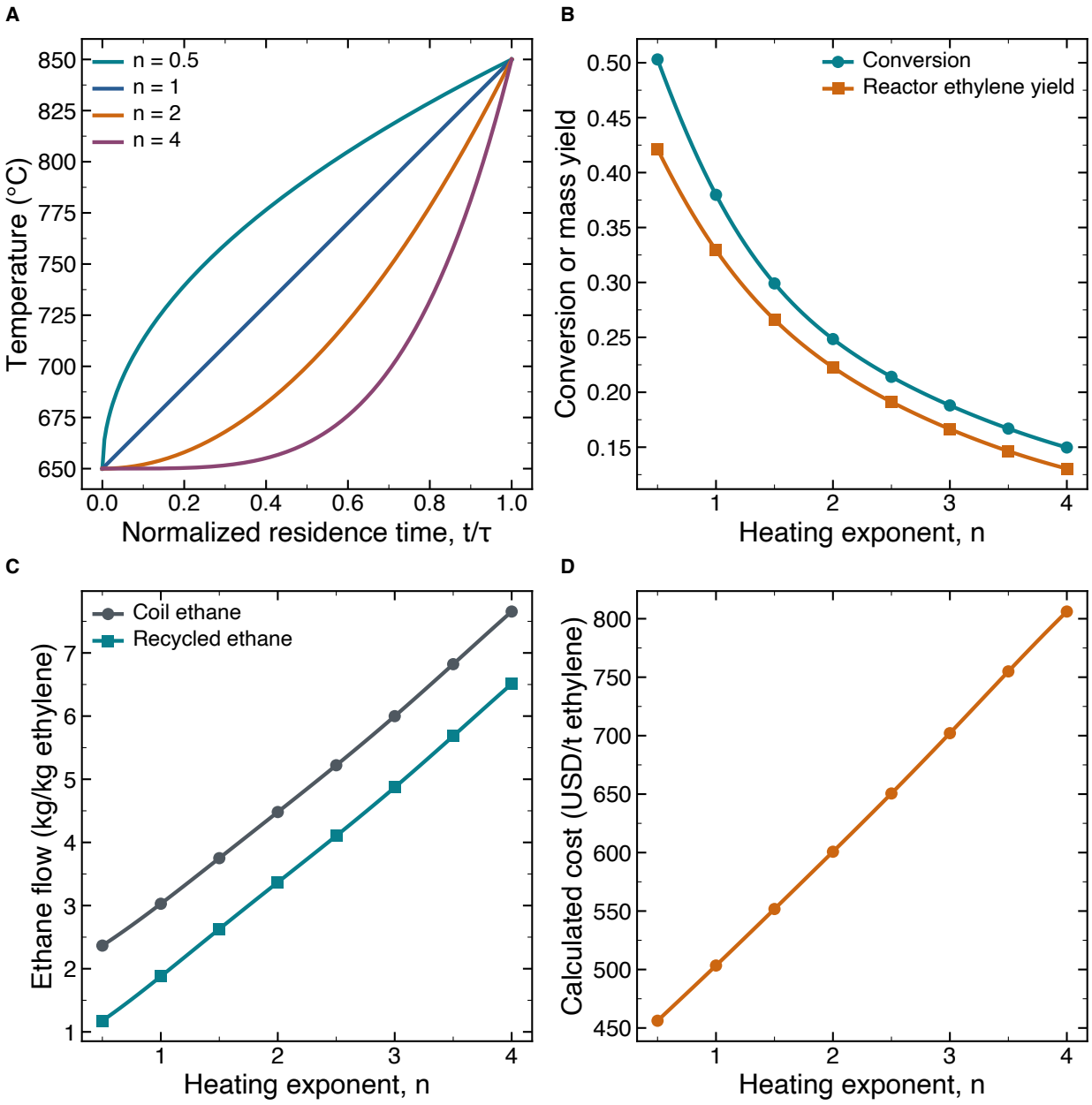


Figure S1. Heating-history effects at fixed outlet conditions. **(A)** Prescribed profiles from 650 to 850 °C. **(B)** Conversion and reactor ethylene mass yield, **(C)** coil and recycle ethane flows, and **(D)** calculated fired-case cost as the heating exponent changes. Residence time is 0.35 s, steam/ethane is 0.35 kg kg^{-1}, and pressure is 1.5 bar. Curves connect GP evaluations, and symbols identify selected evaluations. All process and price assumptions are fixed.

S7. Browser application

Figure S2 shows the released application running the reference fired case. The interface groups reactor inputs, the process flowsheet, and calculated cost and carbon outputs in one view.

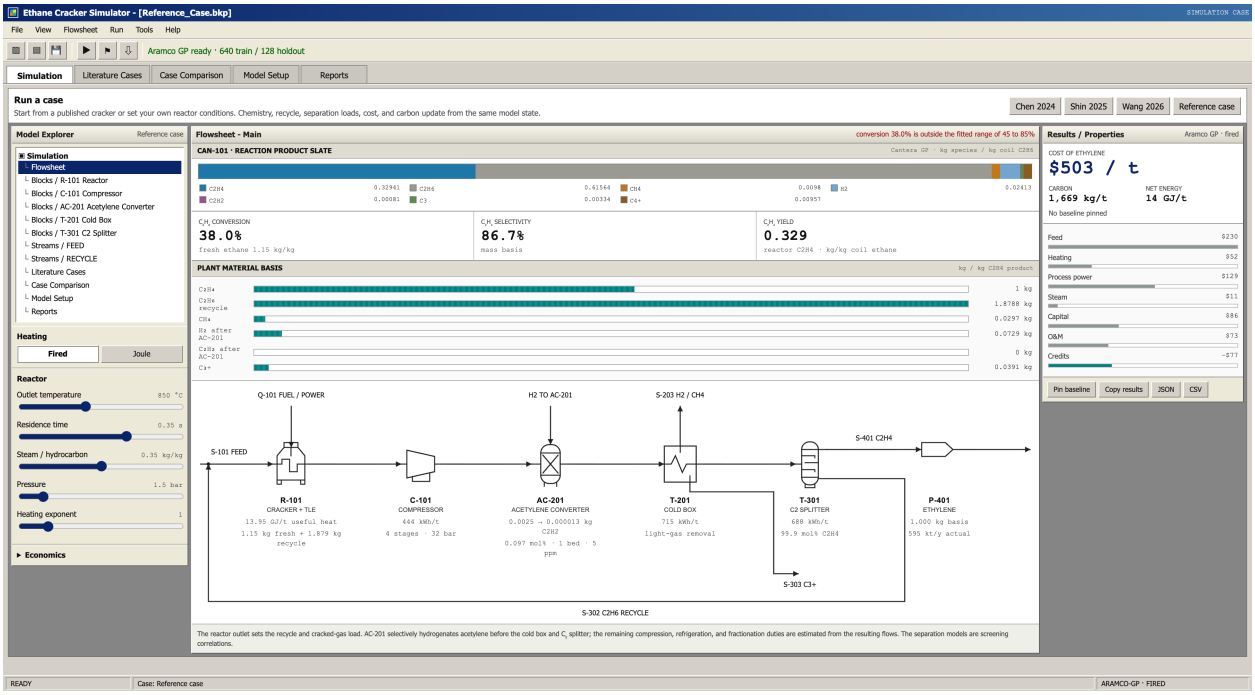


Figure S2. Browser application running the reference fired case. Inputs are 850 °C, 0.35 s, steam/ethane mass ratio 0.35, 1.5 bar, and heating exponent 1. The application includes the AC-201 acetylene converter and reports 503 USD t^{-1} ethylene, consistent with the main-text reference case.